Session 9 Normal Distribution

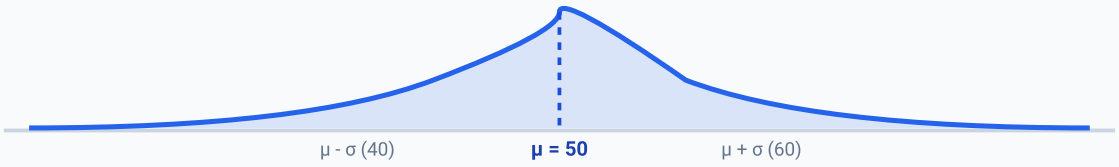
Gaussian Distributions, Empirical Rules & Continuous Random Variables

1. Simulating Normal Distribution & Bell Curve Plotting

Task: Generate 100 random numbers sampled from N(μ = 50, σ = 10) using numpy and plot the bell curve.

```
import numpy as np import matplotlib.pyplot as plt np.random.seed(42) data = np.random.normal(loc=50, scale=10, size=100) # Plot histogram and Gaussian curve plt.hist(data, bins=15, density=True, alpha=0.6, color='#2563eb', edgecolor='white') plt.title("Normal Distribution (μ=50, σ=10)") plt.xlabel("Value") plt.ylabel("Probability Density") plt.show()
```

BELL CURVE VISUALIZATION (GAUSSIAN SHAPE)



SAMPLE METRICS

Sample Mean: **49.04** | Sample Std Dev: **9.08** (Approximating true parameters μ=50, σ=10)

2. Fitness Tracker Daily Step Counts Symmetry Check

Dataset: 20 daily step measurements = [7800, 8100, 8200, 8000, 7900, 8300, 8000, 8100, 8200, 8000, 7900, 8100, 8200, 8000, 8400, 7800, 8000, 8100, 7900, 8200]

```
import numpy as np from scipy import stats steps = np.array([7800, 8100, 8200, 8000, 7900, 8300, 8000, 8100, 8200, 8000, 7900, 8100, 8200, 8000, 8400, 7800, 8000, 8100, 7900, 8200]) mean_val = np.mean(steps) median_val = np.median(steps) mode_val = stats.mode(steps, keepdims=True).mode[0] print(f"Mean: {mean_val:.2f} | Median: {median_val:.2f} | Mode: {mode_val}")
```

SYMMETRY VERIFICATION

- **Mean: 8,060.00 | Median: 8,050.00 | Mode: 8,000.00**
- **Conclusion:** The three central tendency measures are approximately equal (Mean ≈ Median ≈ Mode), confirming that daily step counts exhibit the symmetric, bell-shaped properties expected of a normal distribution.

3. Gaming Score Simulation & The 68-95-99.7 Empirical Rule

Scenario: Simulating 200 player scores in PUBG sampled from N(μ = 70, σ = 15).

```
import numpy as np np.random.seed(101) scores = np.random.normal(loc=70, scale=15, size=200) # Count players within 1 standard deviation [μ - σ, μ + σ] -> [55, 85] within_1_sd = np.sum((scores >= 55) & (scores <= 85)) percentage = (within_1_sd / 200) * 100 print(f"Players within 1 SD (55 to 85): {within_1_sd} / 200 ({percentage:.1f}%)")
```

Simulation Result

138 Players (69.0%)

Scored between 55 and 85 points

Theoretical 68-95-99.7 Rule

~136 Players (68.27%)

Expected theoretical value for N=200

EMPIRICAL CHECK

The simulated count of **138 players (69.0%)** closely matches the theoretical 68.27% expected under the 68-95-99.7 rule for normal distributions, validating the experimental sample.

4. Real-World Application: Zomato Food Delivery Times

Why Normal Distribution Fits Zomato Delivery Times

Zomato delivery times naturally follow a normal distribution centered around an average (e.g., 30 minutes). According to the **Central Limit Theorem**, total delivery time is the sum of multiple independent random variables—such as kitchen preparation time, traffic delay, and distance traveled. Minor positive and negative variations across these steps aggregate into a symmetric bell curve where typical orders cluster near the mean, while extreme fast or slow deliveries fade into thin tails.